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DIRECT FILE SHARING APPLICATION

Network Programming Project — Topic Identification

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Network Programming Project

ONE LOVE. ONE FUTURE.

1. Problem Definition & Core Use Cases

Core Use Cases

- **UC1 (Account Creation)**: Create user account and authenticate.
- **UC2 (Publish Files)**: Announce local files in the catalog.
- **UC3 (Cancel Publication)**: Withdraw files from the catalog.
- **UC4 (Search by Name)**: Search files and view hosting machines.
- **UC5 (Download File)**: Select a machine and download directly.
- **UC6 (Report Error)**: Report failed downloads for catalog cleanup.

System Actors

File Publisher: Makes local files available for sharing.
File Seeker: Searches and downloads files directly.

Problem Definition

- **Resource Discovery & Access**: Locate which active machines hold desired files, access authorized shares, and maintain an accurate catalog as machines join and leave.
- **Direct Peer-to-Peer Transfer**: Establish direct machine-to-machine connections to stream file data reliably and with integrity, without any intermediary storage server.

2. Inputs & Outputs Specification

Account & Catalog Operations

- **Account Creation & Login**:
  - *In:* Username, password credentials.
  - *Out:* Auth status (success/failure), session state.
- **File Publication**:
  - *In:* File name, metadata, host machine address.
  - *Out:* Publication confirmation; catalog index updated.
- **Cancel Publication**:
  - *In:* Target file identifier to revoke.
  - *Out:* Revocation status; file removed from catalog.

Search, Download & Maintenance

- **Search for Files**:
  - *In:* Target file name / keyword query.
  - *Out:* List of machine addresses hosting matching files.
- **Direct File Download**:
  - *In:* Selected machine address, target file name.
  - *Out:* File content streamed to local storage.
- **Report Indexing Error**:
  - *In:* Unsuccessful download signal (failed host/file).
  - *Out:* Error logged; catalog updated to purge entry.

3. Functional & Non-Functional Requirements

Functional Requirements (FR)

- **FR1 (Account Management)**: Support user account creation and login.
- **FR2 (File Publishing)**: Enable users to publish available files and cancel publications.
- **FR3 (Search & Lookup)**: Search files by name; return addresses of machines holding those files.
- **FR4 (Direct Download)**: Allow users to retrieve files directly from a designated machine.
- **FR5 (Catalog Self-Correction)**: Receive error reports for failed downloads and update file catalog.

Non-Functional Requirements (NFR)

- **Data Integrity**: Files transferred directly between machines must remain bit-for-bit identical to the source.
- **Catalog Accuracy**: The file index must accurately reflect available files and prune invalid entries.
- **Fault Resilience**: Transfer failures or offline machines must be handled gracefully with clear user feedback.
- **Controlled Privacy**: Only files explicitly published by the user are accessible to other network peers.



THANK YOU !